



AS5041 – Final Project Presentation

GPU Accelerated Reduced Order Modeling and Analysis of NACA0012 Airfoil

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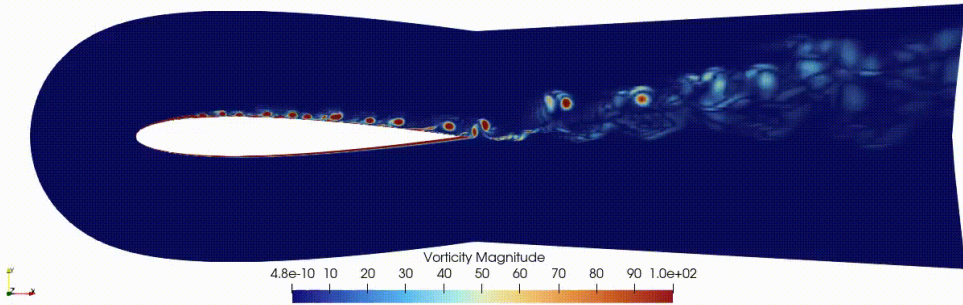
AE21B053

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MOTIVATION



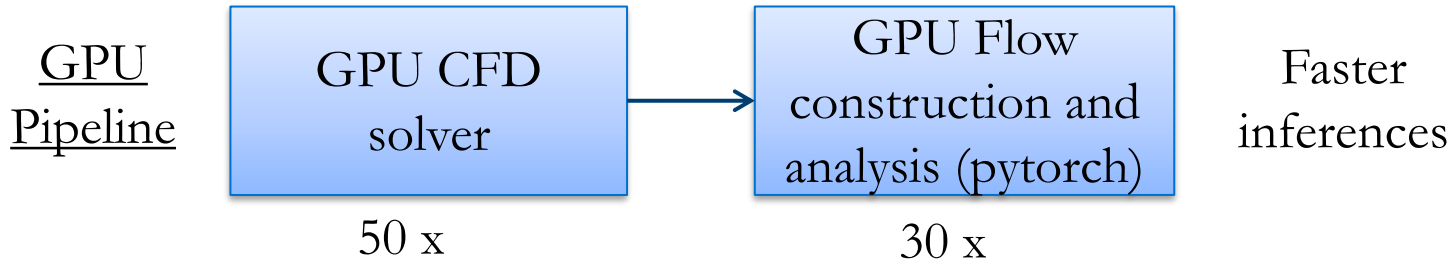
- Computational demands of scale resolving high-fidelity CFD simulations are increasing rapidly.



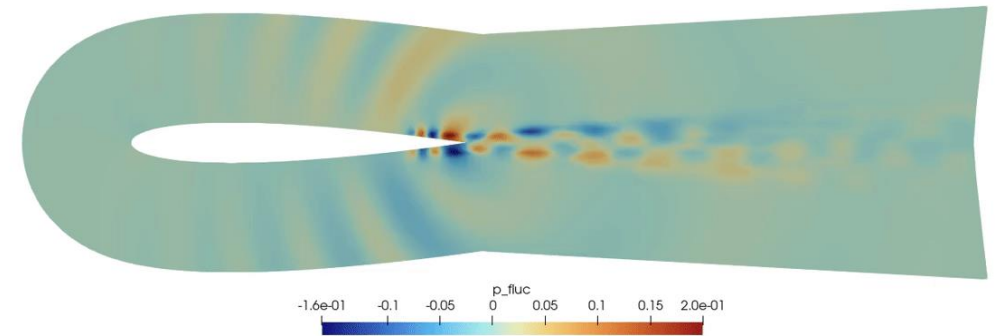
Vorticity contour NACA0012 M=0.3 6 AOA
~0.9M to resolve the flow

- POD and DMD - extract dominant flow features , enable faster flow reconstruction, analysis, and physical insight.
- Involves heavy matrix operations, computational bottleneck on CPUs

- GPUs can do these operations in an efficient faster way



- To use this GPU-parallelized pipeline to analyze trailing edge noise in NACA0012 airfoil



Trailing edge Noise in NACA0012 M=0.5

- Governing equation and Data generation
- Proper Orthogonal Decomposition(POD) Analysis
- Total Least Square Dynamic Mode Decomposition(tls-dmd) Analysis
- Trailing edge noise analysis at $M=0.5$
- Trailing edge noise analysis at $M=0.3$ and comparison
- Conclusion

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Solver : COMP-SQUARE (*Vadlamani et al. (2018)*)

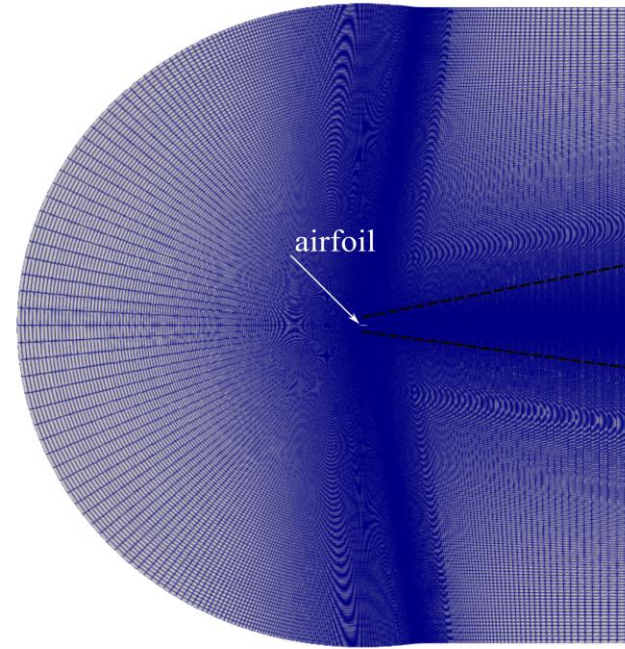
- Finite Difference Unsteady compressible N-S Solver
- Validated on several subsonic and supersonic test cases

$$\frac{\partial}{\partial t} \left(\frac{U}{J} \right) = \frac{\partial}{\partial \xi} \left(\frac{1}{Re} F_v - F_I \right) + \frac{\partial}{\partial \eta} \left(\frac{1}{Re} G_v - G_I \right) + \frac{\partial}{\partial \zeta} \left(\frac{1}{Re} H_v - H_I \right)$$

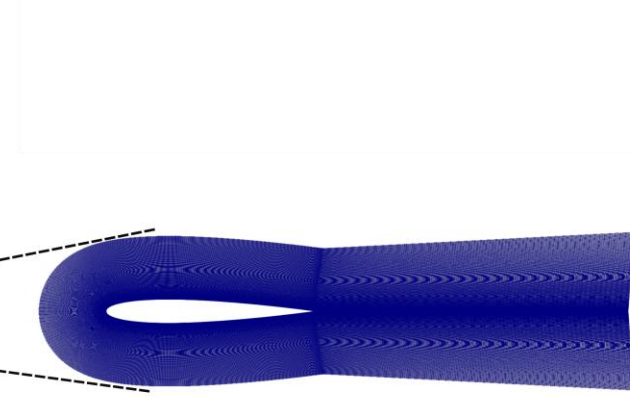
$$U_{con} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E_t \end{bmatrix} \begin{array}{l} \eta, \xi, \zeta - \text{Curvilinear coordinate system} \\ F_I, G_I, H_I - \text{Inviscid flux vectors} \\ F_v, G_v, H_v - \text{Viscous flux vectors} \\ J - \text{Jacobian of the transformation matrix} \end{array}$$

Cases:

- NACA0012 airfoil : $M=0.5$ 0° AOA
- NACA0012 airfoil : $M=0.3$ 0° AOA



(a)
Computational Grid(0.8M)
(Nx,Ny) = (1126,727)



(b)
Analysis Grid(N=0.33M)
(Nx,Ny) = (926,363)

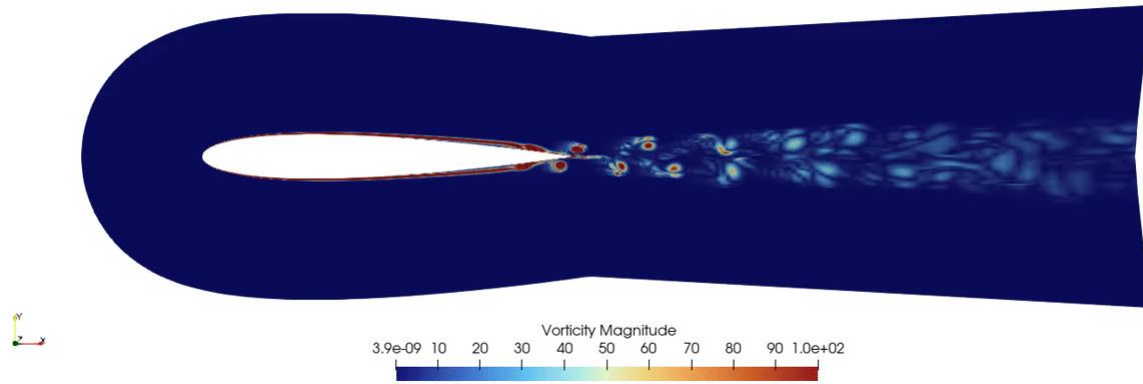
Q – Snapshot matrix $\sim 1\text{GB}$

$$Q = [\rho \ \rho u \ \rho v \ P]^T \in R^{4N \times 400} \text{ (Yoimi et al, JFM)}$$

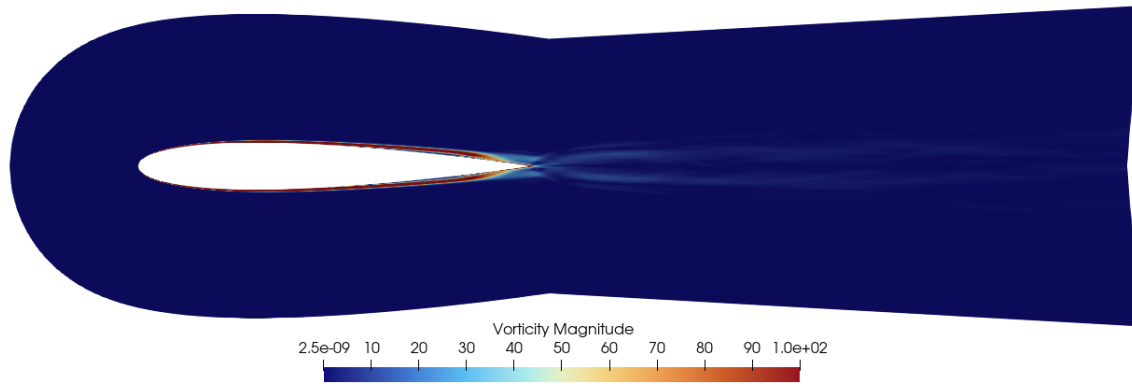
Sampling rate = $1/0.0125$

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PROPER ORTHOGONAL DECOMPOSITION(POD)

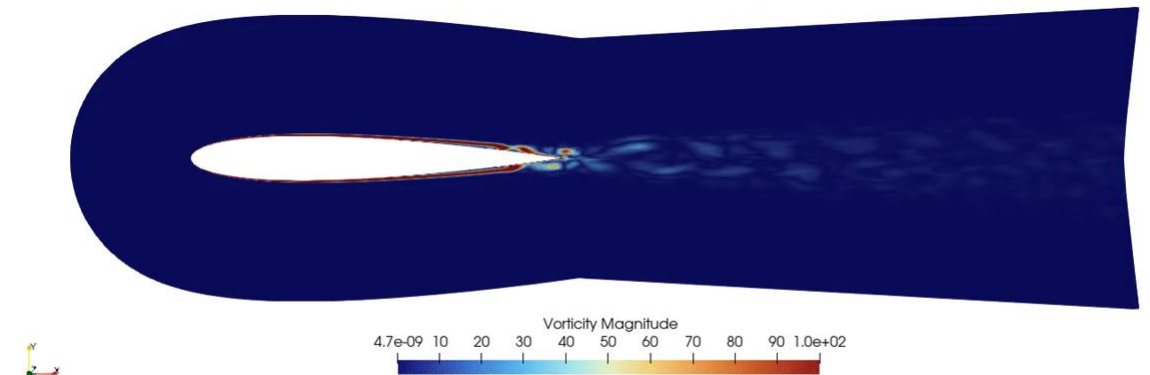
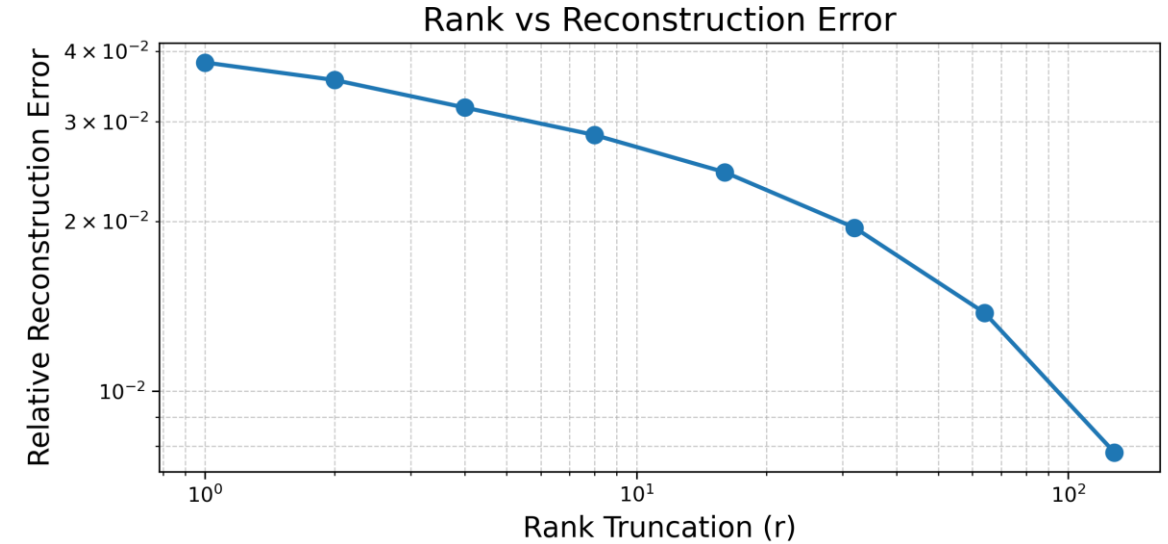


Vorticity contour of FOM: M=0.5 0AOA



Vorticity contour of dominant mode: M=0.5 0AOA

- Dominant mode captures $\sim 99.8\%$ energy



Vorticity contour of ROM($r=4$, 99.9%): M=0.5 0AOA

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TOTAL LEAST SQUARE DMD



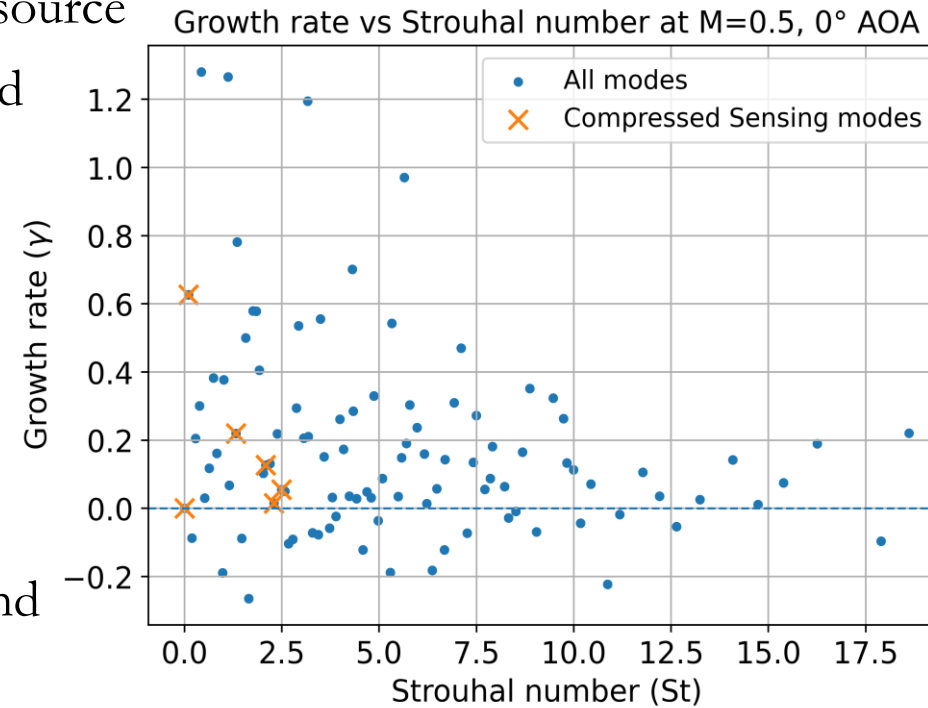
- Tls-dmd is employed to reveal the trailing edge noise source
- Compressed sensing (Mode selection algorithm) is used (*Ohmichi et al 2017*)
- Iterative Greedy method

$\tilde{X} \approx \tilde{\Phi} D_{\alpha} V_{\text{and}}$ $\longrightarrow$ Reconstructed using DMD modes

minimize $\|\alpha\|_0$ subj. to $J(\alpha) \leq \epsilon$,

$$J(\alpha) = \|\tilde{X} - \tilde{\Phi} D_{\alpha} V_{\text{and}}\|_2 \quad \|\alpha\|_0 = K$$

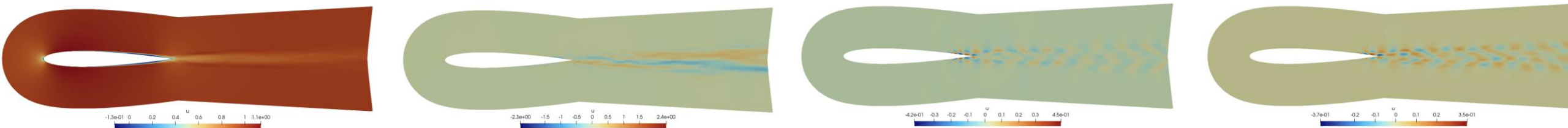
- K=10 modes are chosen out of all modes from tls-dmd



$$\gamma_i = \frac{\text{Re}(\log(\lambda_i))}{\Delta t_c}$$

$$St_i = \frac{\text{Im}(\log(\lambda_i))}{2\pi U_{\infty} \Delta t_c}$$

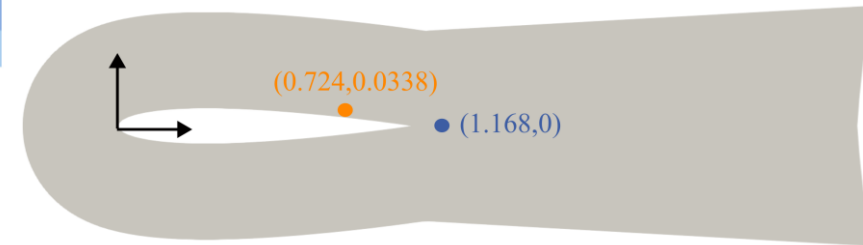
λ_i - eigenvalues of dmd modes



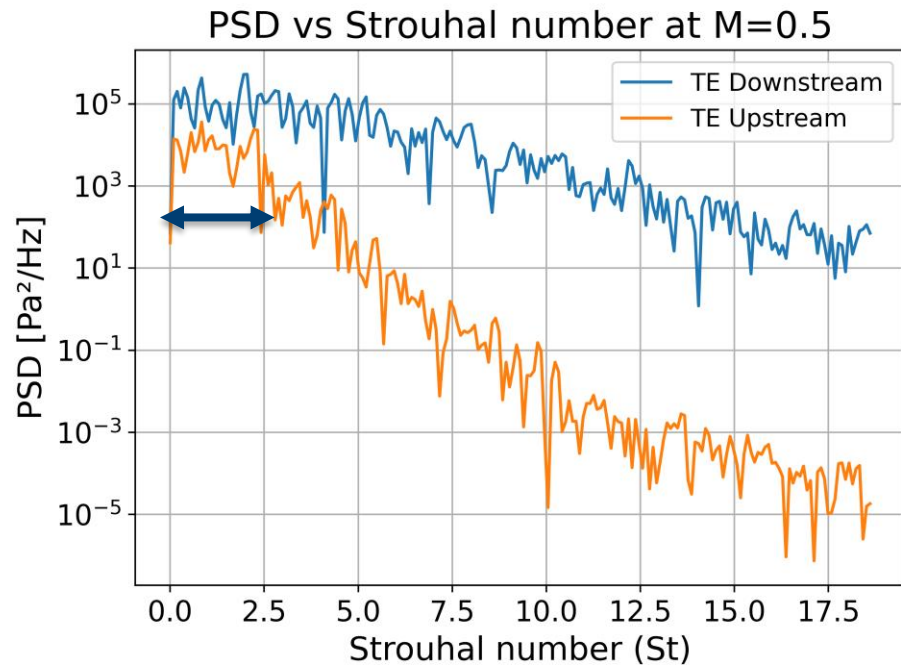
U-velocity contour: (a) St = 0.0 (b) St = 0.1002 (c) St = 2.295 (d) St = 2.085

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TRAILING EDGE NOISE at M=0.5



Probe locations



- Power Spectral Density(PSD) of two locations are analysed
- Compressed sensing is able to extract characteristic frequencies



Pressure fluctuation contour: (a) $St = 0.1$ (Dipole source)

(b) $St = 2.295$ (Quadrupole source)

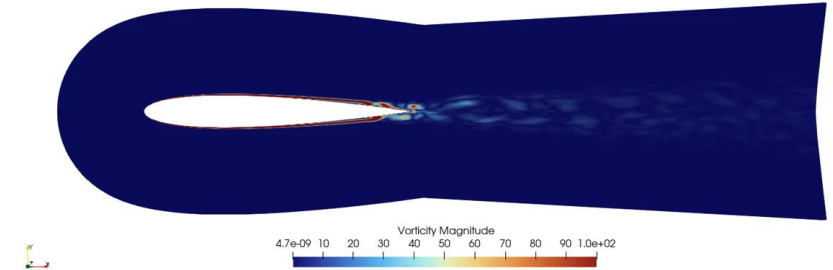
- Interaction between the vortex and the trailing edge leads to dipole sound source
- Interaction between vortices from both sides leads to quadrupole sound source
- Dipole – low frequency, Quadrupole – high frequency

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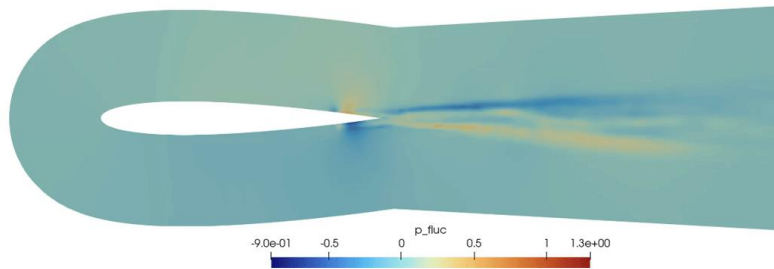
CONCLUSION



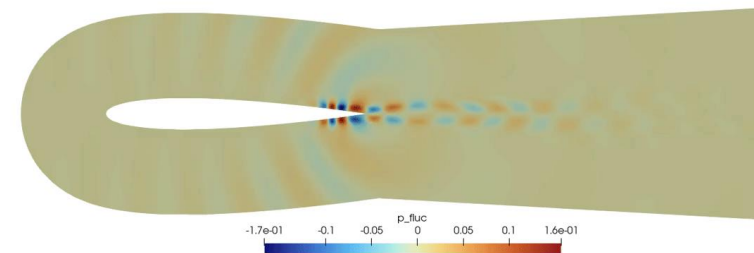
- POD analysis reveals that a single dominant mode captures about 99.8% of energy
- Compressed sensing algorithm captures all main tonal frequencies
- Dipole and Quadrupole noise sources are identified from tls-dmd modes
- Dipole – low frequency noise , Quadrupole – high frequency noise



Vorticity contour of ROM($r=4$, 99.9%)

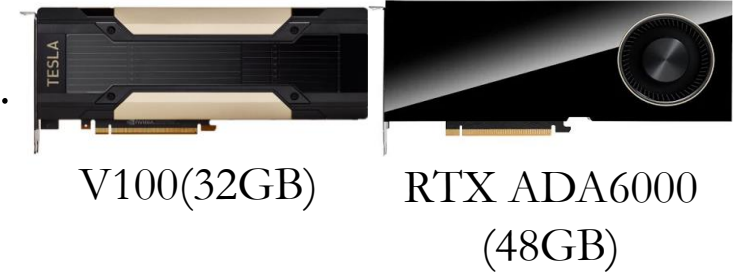


Pressure fluctuation contour: (a) $St = 0.1$ (Dipole source)



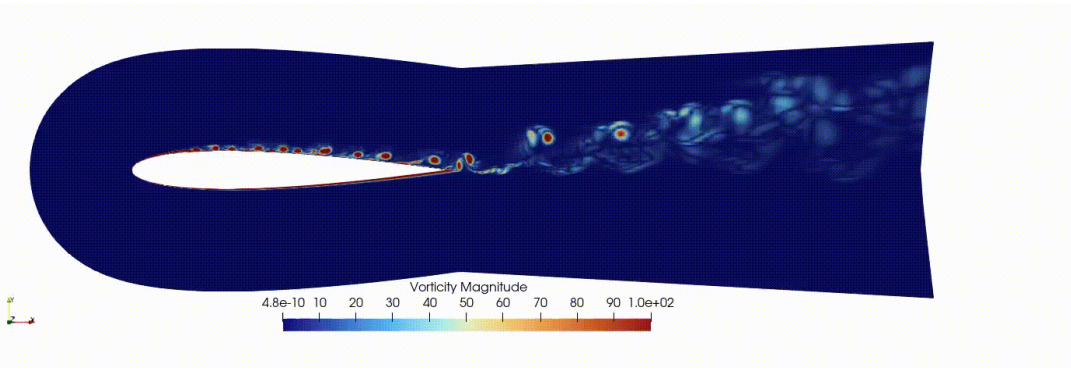
(b) $St = 2.295$ (Quadrupole source)

- Thanks to **Mr. Abdul** for solver access and technical discussions.
- Thanks to **Prof. Nagabhushana Rao Vadlamani** for computational resources.
- **ChatGPT** for assistance with PyTorch and compressed sensing.

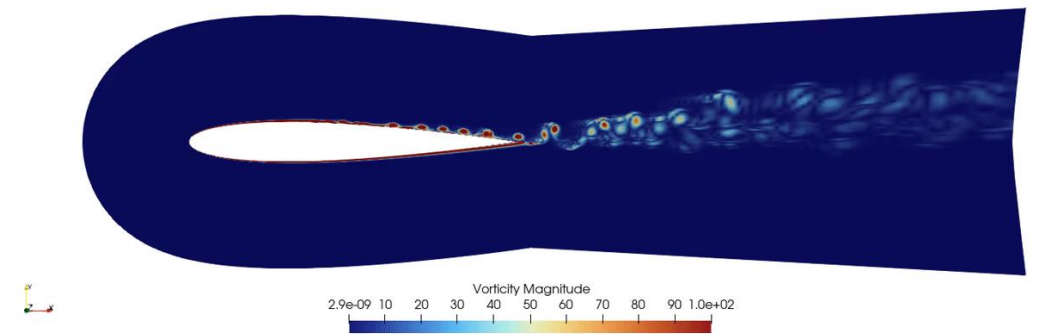


References

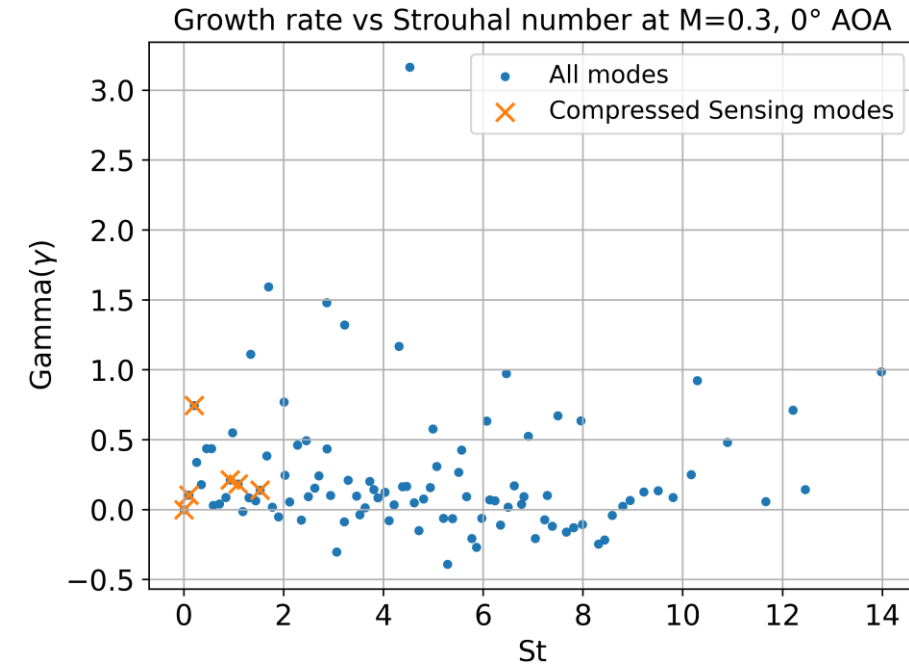
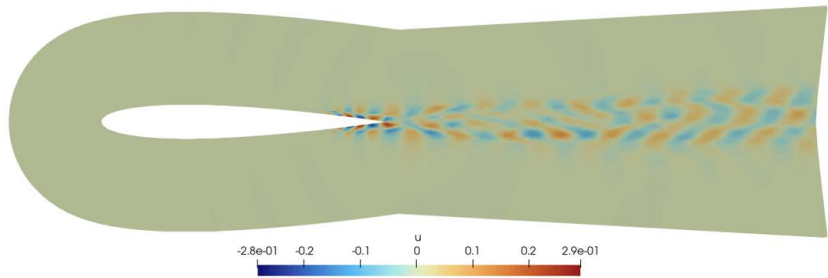
1. Vadlamani, Nagabhushana Rao, Paul G. Tucker, and Paul Durbin. *Distributed Roughness effects on transitional and turbulent boundary layers*. Flow, Turbulence and Combustion 100, no. 3 (2018): 627-649.
2. OHMICHI, Y. 2017 *Preconditioned dynamic mode decomposition and mode selection algorithms for large datasets using incremental proper orthogonal decomposition*. AIP Adv. 7 (7), 075318.
3. Yoimi Kojima et al, *On the origin of quadrupole sound from a two-dimensional aerofoil trailing edge*, JFM, 2023



Thank You



SVD Speedup over AMD EPYC CPU	V100	RTX ADA6000
Double Precision	14.72 x	31.17 x
Single Precision	16.16 x	40.17 x



- Dipole frequency decreases and Quadrupole frequency increases with Mach number